

Digital Circuits Homework 2

Spring 2026

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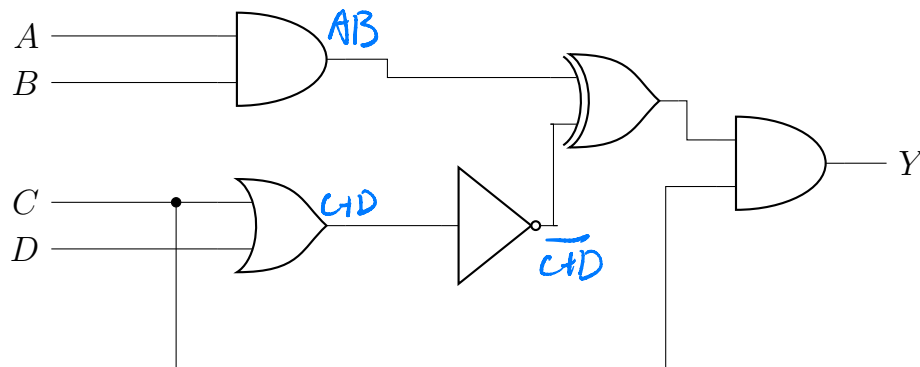
Due: 23:59, April 10, 2026

Instructions

- Write your answers neatly in the space provided.
- Show all necessary steps for your derivations.
- Late submission may result in a reduction in score.
- Total: 100 pts

Problem 1. (15 pts)

Write the Boolean expression for the following circuit and simplify it to its minimal Sum of Products (SOP) form.



$$[(A \cdot B) \oplus (\overline{C+D})] \cdot C$$

$$= [(A \cdot B) \oplus (\bar{C} \cdot \bar{D})] \cdot C$$

$$= [(A \cdot B \cdot (C+D)) + (A \cdot \bar{B} \cdot \bar{C} \cdot \bar{D})] \cdot C$$

$$= ABC + ABCD = ABC$$

Problem 2. (20 pts)

Apply DeMorgan's theorems to simplify the following Boolean expressions. Write down results in the SOP form.

$$(1) X = \overline{(A + \overline{B})(C + D)}$$

$$(2) X = \overline{A\overline{B} + (C + \overline{D})E + \overline{F}}$$

$$1) X = \overline{(A + \overline{B})(C + D)} = \overline{A}B + \overline{C}\overline{D}$$

$$(2) X = \overline{A\overline{B} + (C + \overline{D})E + \overline{F}}$$

$$= (\overline{A} + B) [\overline{(C + \overline{D}) + E}] \cdot F$$

$$= (\overline{A} + B) [\overline{C}\overline{D} + \overline{E}] \cdot F$$

$$= \overline{A}\overline{C}\overline{D}F + B\overline{C}\overline{D}F + \overline{A}\overline{E}F + B\overline{E}F$$

Problem 3. (20 pts)

Convert the following Boolean functions to standard Sum of Products (SOP) form and list their minterms (assuming variables A, B, C , use Boolean algebra):

(1) $F(A, B, C) = AB$

(2) $F(A, B, C) = A\bar{B}C + \bar{A}BC + AC$

11)

$AB \backslash C$	0	1
00	0	0
01	0	0
11	1	1
10	0	0

$$F(A, B, C) = ABC + AB\bar{C}$$

12)

$AB \backslash C$	0	1
00	0	0
01	0	1
11	0	1
10	0	1

$$F(A, B, C) = \bar{A}BC + ABC + A\bar{B}C$$

Problem 4. (15 pts)

Minimize the following function to its simplest SOP form using a Karnaugh map. Draw the corresponding K-map and clearly show your groupings.

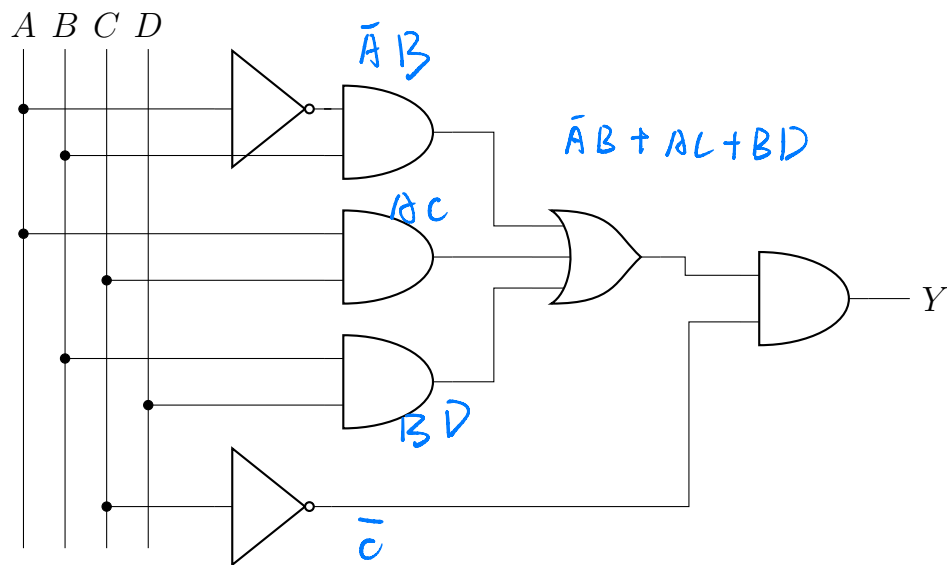
$$F(A, B, C, D) = \sum m(0, 2, 3, 5, 7, 8, 10, 11, 14)$$

			0	1	3	2
			00	01	11	10
	AB \ CD		1	0	1	1
0	00					
4	01		0	1	1	0
12	11		0	0	0	1
8	10		1	0	1	1

$$F(A, B, C, D) = \bar{B}\bar{D} + \bar{A}BD + \bar{B}C + AC\bar{D}$$

Problem 5. (15 pts)

Find the Boolean expression for the circuit and construct its complete truth table.



$$Y = (\bar{A}B + AC + BD) \cdot \bar{C} = \bar{A}B\bar{C} + BD\bar{C} = B\bar{C}(\bar{A} + D)$$

A	B	C	D	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

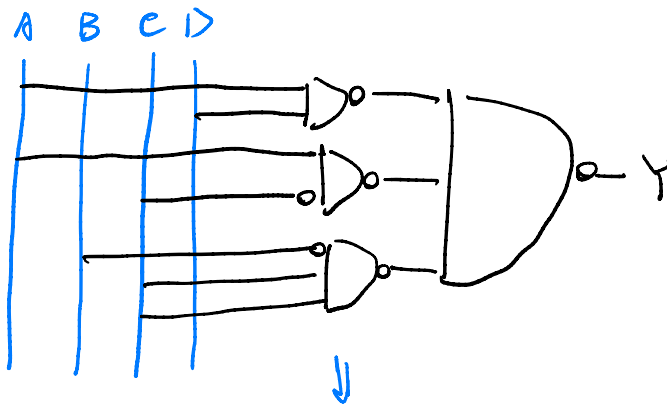
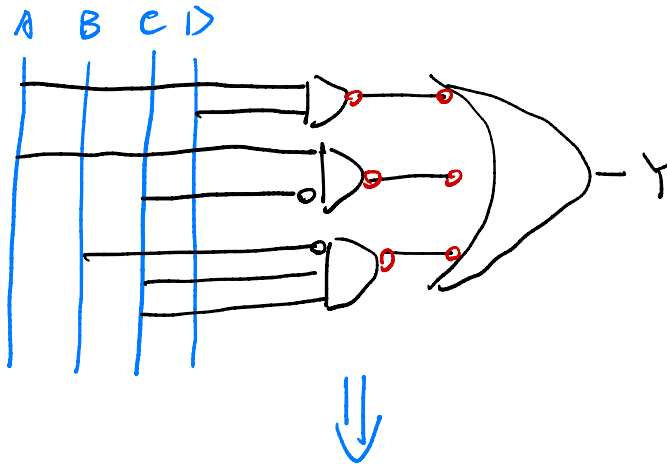
Problem 6. (15 pts)

Implement the following Boolean function using **at most** two 2-input NAND gates and two 3-input NAND gates. Draw the schematic of your mapped logic circuit. **Assume** complemented inputs are available.

$$Y = (A + B'C)(D + AC')$$

$$= AD + A\bar{C} + \bar{B}CD$$

$\overline{AB}^{\overline{CD}}$	00	01	11	10
00	0	0	1	0
01	0	0	0	0
11	1	1	1	0
10	1	1	1	0



$$Y = \text{NAND}(\text{NAND}(A, D), \text{NAND}(A, \bar{C}), \text{NAND}(\bar{B}, C, D))$$